

TECHNICAL & FINANCIAL DUE DILIGENCE

Infrastructure & Compute Forecast

What the cloud costs per site, where the platform changes gear,
and why infrastructure never threatens the software margin.

Document

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Prepared

2 August 2026

Basis

Azure list rates, East US

Status

Model · not yet metered

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1. What this answers

Infrastructure costs **\$0.72–\$0.91 per site per month** against \$22–38 of recurring revenue, stays under 2% of ARR at every stage of the plan, and falls as the estate grows.

COST PER SITE \$0.72 per month at 8,000 sites	SHARE OF ARR 1.8% never exceeds 2.0%	YEAR-5 SPEND \$68.8K annual, 8,000 sites	FIRST HARD GATE 250 sites — Postgres leaves Burstable
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Three questions get asked of this platform, in this order. What does the cloud actually cost per site, given the recurring price is \$22–38 and infrastructure has to disappear inside it. At what site counts does each service change tier, so capacity is planned rather than discovered. And does the cost curve stay under the revenue curve — the question behind the other two.

2. Telemetry assumptions

Every figure in this document derives from these six numbers. Change one and the model moves.

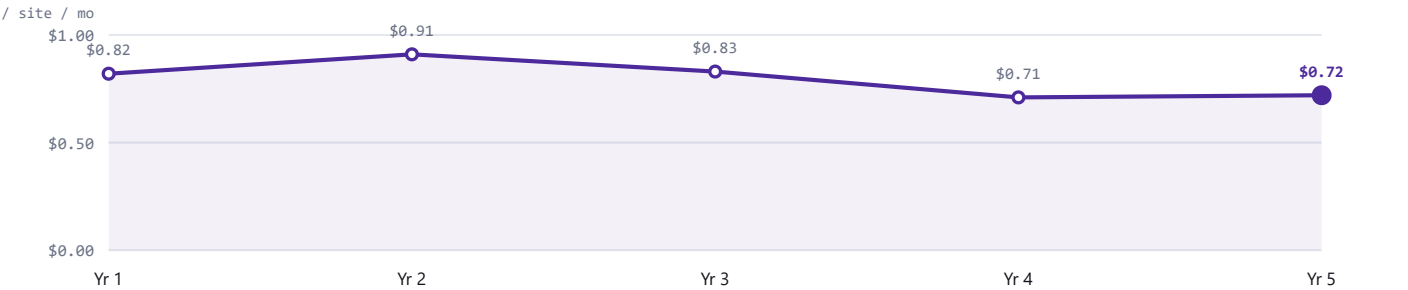
ASSUMPTION	VALUE	RATIONALE
Metrics per site	8	Matches the tile count on both configured site types
Sample interval	60 s	Below the fastest silence-detection window, so a dead sensor is caught inside one sweep
Messages per site / day	1,440	Batched — one message carrying all eight metrics
Readings per site / day	11,520	8 metrics × 1,440 messages
Bytes per reading	48	Narrow row: device key, metric, value, timestamp
Full-resolution retention	90 d	In Postgres, then compressed to blob at ~12% of raw

Batching is the most consequential decision here. IoT Hub bills per message under 4 KB, so one message per metric would multiply that line **eightfold** for identical data — the difference between S1 ×7 and S2 ×3 at 1,800 sites.

3. Cost per site

Priced at pay-as-you-go list rates. No reserved instances, no enterprise agreement, no negotiated discount — each of which is available later and only improves these figures.

Monthly infrastructure cost per site



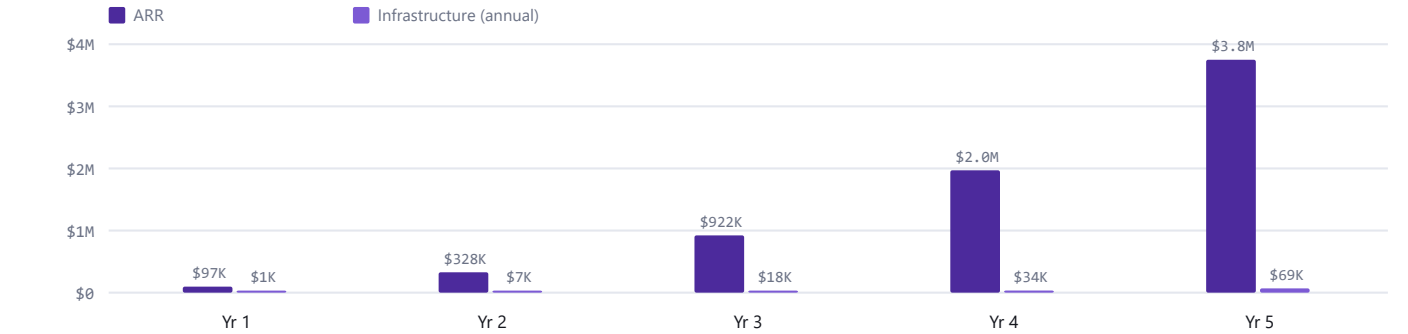
Cost per site falls as the estate grows: the fixed floor — database compute, observability baseline, Key Vault — amortises across more sites. There is no point at which this inverts.

YR	SITES	MSG/DAY	IOT HUB	POSTGRESQL	FUNCTIONS	LOG	ANALYTICS	APIM	EGRESS	TOTAL/MO	\$/SITE
1	150	216K	S1 × 1	B2s	\$13		\$35	\$11	\$3	\$122	\$0.82
2	600	864K	S1 × 3	D2ds_v5	\$72		\$159	\$53	\$38	\$544	\$0.91
3	1,800	2592K	S1 × 7	D4ds_v5	\$227		\$490	\$167	\$132	\$1K	\$0.83
4	4,000	5760K	S2 × 1	D4ds_v5	\$512		\$1K	\$375	\$305	\$3K	\$0.71
5	8,000	11520K	S2 × 2	D8ds_v5	\$1K		\$2K	\$753	\$618	\$6K	\$0.72

Monthly figures. IoT Hub and PostgreSQL tiers shown as provisioned; all other lines are consumption-priced.

4. Cost against revenue

Annual recurring revenue against annual infrastructure cost



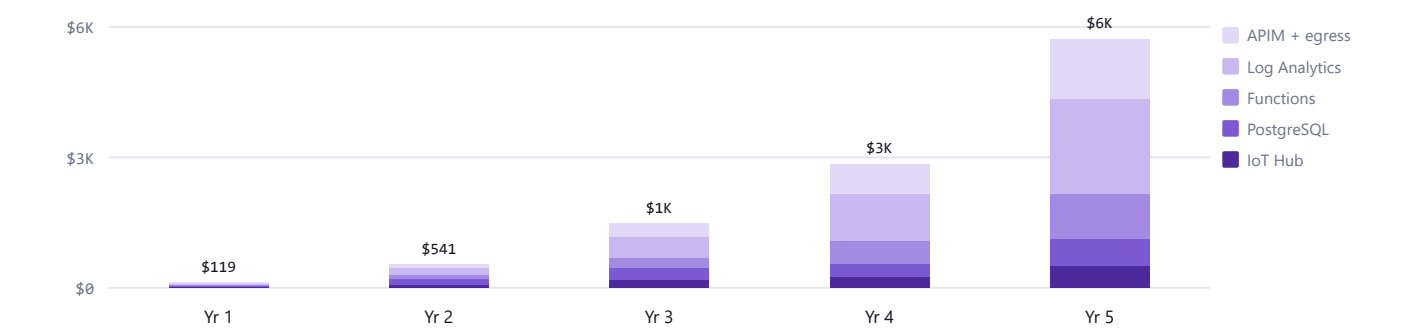
Drawn to a shared axis. Infrastructure is visible only as a hairline at this scale, which is the point.

YEAR	INFRASTRUCTURE / YR	ARR	INFRA AS % OF ARR
Year 1	\$1K	\$97K	1.5%
Year 2	\$7K	\$328K	2.0%
Year 3	\$18K	\$922K	1.9%
Year 4	\$34K	\$2.0M	1.7%
Year 5	\$69K	\$3.8M	1.8%

The 82% software gross margin assumed in the business plan is not at risk from cloud spend. It is at risk from support and field cost, which is where scrutiny belongs.

5. Where the money goes

Monthly cost composition



Two lines dominate, and neither is the one intuition suggests.

LARGEST BY YEAR 3

Log Analytics

Observability, not compute — \$490/mo rising to \$2,200. It is also the most compressible: moving verbose diagnostics to a Basic table roughly halves it. The first optimisation to make, not the first thing to buy more of.

LEAST ELASTIC

PostgreSQL

The largest fixed commitment. Everything else scales smoothly with usage; the database steps, and each step needs planning ahead of the load rather than in response to it.

CHEAPER THAN EXPECTED

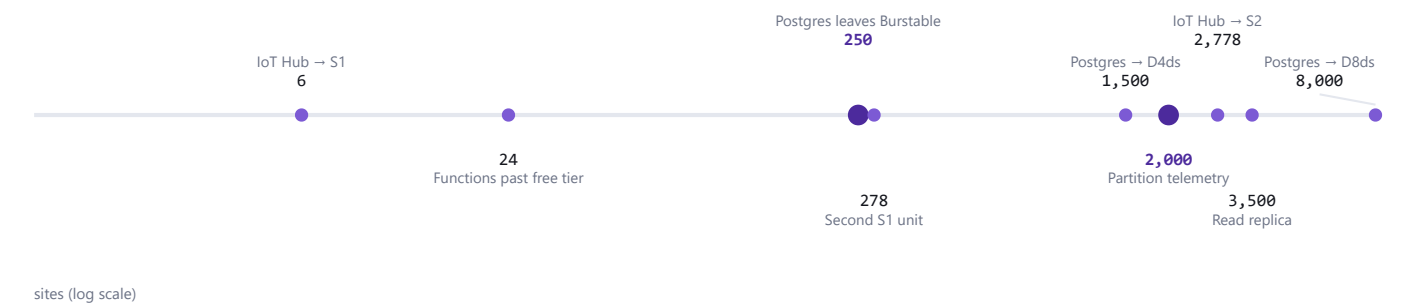
IoT Hub

\$500/mo at 8,000 sites, precisely because telemetry is batched. This is the line that would have dominated under a naive per-metric design.

6. Scaling inflection points

Site counts at which something must change. These are planning triggers, not failures.

When the platform changes gear



Log scale. Marked points need a decision; the rest are automatic.

SITES	WHAT CHANGES	ACTION REQUIRED
~6	IoT Hub leaves the free tier	None — \$25/mo
~24	Functions exceeds 1M free monthly executions	None — marginal cost is pennies
~250	PostgreSQL must leave Burstable	First real decision. B-series runs on CPU credits; sustained ingest exhausts them and latency collapses without warning. Migrate before this point.
~278	IoT Hub needs a second S1 unit	Additive and instant; no migration
~1,500	PostgreSQL → D4ds_v5	Index maintenance on telemetry becomes the bottleneck ahead of raw CPU

SITES	WHAT CHANGES	ACTION REQUIRED
~2,000	Partition telemetry by month	Vacuum and scan cost on one large table starts to hurt. Partitioning also makes the 90-day retention drop a metadata operation.
~2,778	IoT Hub → S2	Also the point to evaluate consuming Event Hubs directly
~3,500	Read replica for reporting	Analytics begins competing with ingest; a replica is cheaper than sizing the primary for both
~5,000	PostgreSQL → D8ds_v5; streaming archival	Nightly batch archival no longer completes inside its window

The one to plan for is **250 sites**. Everything before it is automatic and everything after it is a scheduled migration with a known cost — but Burstable-to-General-Purpose is the only transition where mistiming produces an outage rather than a bill.

7. Exclusions

Stated plainly, because a forecast that omits its own gaps is not usable.

Non-production environments

Dev and staging add roughly 25–30% in years 1–2, less thereafter as they stay small while production grows.

Hub-side compute

The on-premises appliance is customer hardware, costed inside the \$1,800 site kit rather than here.

AI inference

Tier 3 prescriptive analytics is design-only. When it ships it becomes the fastest-growing line in this model and needs its own forecast.

Reserved-instance discount

One- and three-year commitments cut database compute 30–55%. Not assumed; available once load is proven.

8. Method

Every figure is produced by `docs/business/infra-forecast.mjs`, committed alongside this document. The assumptions above are arguable, and an argument about them should be settled by changing an input and re-running rather than by debating a spreadsheet nobody can see.

Prices last verified against Azure list rates on 2 August 2026. Re-run before quoting externally.